

MYCRO-TEC®

DIGESTER

Stubble is slow to break down on its own.

Digester adds four selected decomposer organisms to the residue after harvest. They establish at the damp base of the stubble and start working through the cellulose, hemicellulose, lignin and pectin that hold last season's carbon and nutrients.

USED ON CROP RESIDUE IN



VEGETABLES



ORCHARDS



BROADACRE

How residue breaks down

The organisms do the work. Moisture, time and the residue itself set the pace.



01

Establishes at the damp base

The organisms take hold first where residue meets soil and moisture is held, then spread through damp stubble.



02

Opens the structure

Their enzymes and organic acids work on cellulose, hemicellulose, lignin and pectin, so the carbon locked in the fibre becomes available.



03

Turnover builds

The selected organisms and the soil's resident community keep working through the material ahead of the next crop.

What is in it

Trichoderma harzianum FUNGUS	$>2\times10^{8}$ CFU/ML	Fast fungal coloniser of cellulose-rich residue; its enzymes begin opening the fibre.
Aspergillus niger FUNGUS	$>2\times10^{7}$ CFU/ML	Produces organic acids and enzymes that work on the tougher, more lignified material.
Streptomyces lydicus ACTINOMYCETE	$>2\times10^{7}$ CFU/ML	Adds bacterial decomposition alongside the fungi, so breakdown does not rely on one pathway.
Penicillium radicum FUNGUS	$>1\times10^{7}$ CFU/ML	Broadens the decomposer team across the residue surface and the damp base.

Colony-forming units per millilitre at production, as declared on the label. Three fungi and one actinomycete.

Cellulose

The main plant fibre

Hemicellulose

The supporting fibre network

Lignin

The hard, resistant structure

Pectin

The material that binds plant cells

The roles described here are the known behaviour of these organisms on crop residue. They are not a measured result for Digester in a particular paddock or season.

Where it pays off

One application after harvest, where there is residue worth breaking down and moisture to run on.

Residue that will still be there at sowing
Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.

A moisture window you can hit
Applied soon after harvest and close to forecast rain. The breakdown runs on residue moisture.

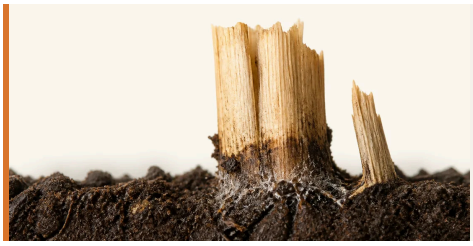
Heavy or lignified loads
Big stubble loads, or material that has sat through a dry summer and stalled.

Where the basics are already in hand
Residue breaks down fastest in a block where water, soil balance and nutrition are being managed. Digester is a step in that program, not a substitute for it.

Hot, dry spells
Activity falls away through prolonged heat and dryness. Line the application up with a viable moisture window.

Time in the tank
Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles.

Nitrogen on its own
Nitrogen feeds whatever biology is already there. It does not add the decomposers. The ammonium sulphate is a starter, not the treatment.



HOW YOU CHECK IT

Digestion starts at the stubble base, where moisture sits. Treated residue pulls away from the soil more easily and the stalks soften and weaken as it runs. That is the one field check a grower can make without a lab.

VEGETABLES

Crop residue after harvest

Foliar onto residue	**1–2 L/ha**
Heavy residue	**up to 3 L/ha**
Ammonium sulphate	**8 kg/ha**
Water	**100 L/ha**
Product : water	**1:50**

ORCHARDS

Prunings and leaf litter on the orchard floor. Not for nut trees.

Foliar onto residue	**1–2 L/ha**
Heavy residue	**up to 3 L/ha**
Ammonium sulphate	**8 kg/ha**
Water	**100 L/ha**
Product : water	**1:50**

BROADACRE

Cereal, canola and legume stubble

Foliar onto residue	**1–2 L/ha**
Heavy residue	**up to 3 L/ha**
Ammonium sulphate	**8 kg/ha**
Water	**100 L/ha**
Product : water	**1:50**

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. Applied once, after harvest, onto the residue. Not for in-crop use. If you run a nutrition program with us, Digester is the post-harvest step and the prescription blend at establishment follows it. If you don't, it stands on its own after harvest.

Rates, mixing and handling

WHAT IS IN IT

CFU/mL at production

$>2\times10^8$	$>2\times10^7$	$>2\times10^7$	$>1\times10^7$
Trichoderma harzianum	*Aspergillus niger*	*Streptomyces lydicus*	*Penicillium radicum*

Colony-forming units per millilitre at production, as declared on the label. Three fungi and one actinomycete.

Application

CROP GROUP	DIGESTER	STARTER	WATER
Vegetables Crop residue after harvest	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
Orchards Prunings and leaf litter on the orchard floor. Not for nut trees.	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
Broadacre Cereal, canola and legume stubble	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. Applied once, after harvest, onto the residue. Not for in-crop use.

Mixing

Shake the drum well. Add Digester to cool, clean, non-chlorinated water first, then add the ammonium sulphate and stir thoroughly.

Strain as you pour. Keep nozzles and filters at 150 µm or larger.

Mix and apply the same day. Left in the tank overnight it grows mycelia and blocks nozzles.

Use only the approved mix of Digester, water and ammonium sulphate. No substitutes for the starter.

Storage

Store at 5–40 °C, out of direct sunlight, drums upright. The lids are vented. Do not freeze.

Twelve months from the date of manufacture. Use opened drums promptly rather than keeping them.

Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

Safety

Keep out of reach of children. May irritate skin or eyes; wash after handling.

Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.

Not certified organic. Allowable-input status is assessed case by case with your certifier.

Talk to us about Digester

Tell us the crop, the residue load and when you expect rain. Your Hybrid-Ag agronomist can advise on timing and rate. Label and safety data sheet available on request.

(03) 5722 7555 **hybridag.com.au**

Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.

